

ImPACT CT Patient Dosimetry Calculator

Version 1.0.4 27/05/2011

Scanner Model:		
Manufacturer:	Siemens	▼
Scanner:	Siemens Sensation 64	▼
kV:	120	▼
Scan Region:	Body	▼
Data Set	MCSET21	
Current Data	MCSET21	
Scan range		
Start Position	-7.5	cm
End Position	44	cm

Organ weighting scheme	ICRP 60	▼
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Acquisition Parameters:		
Tube current	462	mA
Rotation time	0.5	s
Spiral pitch	1.25	
mAs / Rotation	231	mAs
Effective mAs	184.8	mAs
Collimation	19.2 (64* x 0	mm
Rel. CTDI	1.21047	at selected collimation
CTDI (air)	19.543	mGy/100mAs
CTDI (soft tissue)	20.9	mGy/100mAs
$nCTDI_w$	7.0389	mGy/100mAs

CTDI _w	16.3	mGy
CTDI _{vol}	13.0	mGy
DLP	670	mGy.cm

Organ	w _T	H _T (mGy)	w _T .H _T
Gonads	0.2	20	4
Bone Marrow	0.12	8.5	1
Colon	0.12	19	2.3
Lung	0.12	2.7	0.33
Stomach	0.12	19	2.3
Bladder	0.05	21	1.1
Breast	0.05	0.7	0.035
Liver	0.05	17	0.86
Oesophagus (Thymus)	0.05	0.54	0.027
Thyroid	0.05	0.068	0.0034
Skin	0.01	7	0.07
Bone Surface	0.01	12	0.12
Not Applicable	0	0	0
Not Applicable	0	0	0
Remainder	0.05	9.5	0.48
Not Applicable	0	0	0
Total Effective Dose (mSv)			13

Remainder Organs	H _T (mGy)
Adrenals	16
Small Intestine	19
Kidney	21
Pancreas	16
Spleen	17
Thymus	0.54
Uterus	20
Muscle	9.6
Brain	0.0029
Not Applicable	N/A
Not Applicable	N/A
Not Applicable	N/A
Not Applicable	N/A
Other organs of interest	H _T (mGy)
Eye lenses	0.0045
Testes	22
Ovaries	18
Uterus	20
Prostate	21

Scan Description / Comments	
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 Imaging Performance Assessment of CT Scanners, an MHRA Evaluation centre
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